

Frozen Stable Diffusion VAE as a Slice-Wise Encoder for Scalable 3-D Medical Image Generation

Youness Mellak¹, Alexandre Bousse¹, Antoine De Paepe¹,
Dimitris Visvikis¹

Abstract—We address the problem of full-volume three-dimensional (3-D) medical image generation under limited GPU memory. Existing 3-D generative methods are difficult to scale to large clinical volumes, and latent approaches usually require training a dedicated medical autoencoder before generative modeling, which adds substantial cost and complexity. To remove this bottleneck, we propose a latent flow framework that uses a frozen pretrained Stable Diffusion autoencoder (SD-VAE) as a slice-wise encoder for medical volumes. The resulting slice latents are stacked into a 3D latent tensor, and a full-volume flow model is trained directly in this compact space. This design avoids modality-specific autoencoder training while preserving a volumetric generative formulation suitable for large medical images.

Index Terms—3-D medical image generation, flow matching, inverse problems.

I. INTRODUCTION

Clinical medical imaging is inherently volumetric. magnetic resonance imaging (MRI), positron emission tomography (PET), and computed tomography (CT) are acquired and interpreted as full 3-D volumes, and many downstream tasks, including synthesis and inverse problems, depend on modeling that full structure rather than isolated slices. The difficulty is that clinically relevant volumes are large, while memory and compute costs grow quickly with spatial resolution, making direct full-volume generative modeling hard to train on standard single-GPU hardware. Existing approaches address this problem in two main ways. Wavelet-based methods reduce some of the burden but still operate on dense volumetric representations that remain costly at high resolution. Latent diffusion methods are more scalable [1], but they typically require training a dedicated 3-D medical autoencoder for each modality or dataset before generative modeling can begin, making the compression stage itself a major bottleneck [2], [3]. A lighter alternative is to encode axial slices independently with a two-dimensional (2-D) autoencoder and stack the resulting latent maps into a 3-D tensor [4]; however, this approach still requires training a modality-specific encoder from scratch, limiting its scalability across modalities and resolutions. In this work, we address these limitations by using a frozen SD-VAE pretrained on natural images as a ready-to-use slice-wise encoder for medical volumes, then stacking the resulting latent slices into a 3-D latent volume for full-volume generative modeling. This design removes the need for modality-specific autoencoder training entirely and offers a practical path toward memory-efficient volumetric modeling on a single consumer-grade GPU.

¹ LaTIM, INSERM, UMR 1101, *Université de Bretagne Occidentale*, Brest, France.
e-mail: youness.mellak@univ-brest.fr

II. MATERIALS AND METHODS

A. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{D \times H \times W}$ denote a high-resolution 3-D medical volume, where D , H , W are the depth, height, and width dimensions respectively. Our goal is to learn a generative model $p(\mathbf{x})$ without operating directly in voxel space and without training a modality-specific medical autoencoder. Slice-based latent approaches have been proposed to reduce the cost of 3-D generative modeling by encoding axial slices independently [4], but still require a dedicated autoencoder trained on medical data. In this work, we instead exploit a frozen SD-VAE pretrained on natural images as a ready-to-use slice-wise encoder, and show that it transfers effectively to CT, PET, and MRI volumes at clinical resolutions. The resulting latent slices are stacked into a compact 3-D tensor $\mathbf{Z} \in \mathbb{R}^{4 \times D \times 64 \times 64}$, on which a full-volume flow model is trained directly.

B. Pretrained Slice-Wise Latent Representation

Let $\mathbf{x}_d \in \mathbb{R}^{H \times W}$ denote the d -th axial slice of $\mathbf{x}$. Each slice is resampled to resolution $r \times r$ via bilinear interpolation, intensity-normalised to $[-1, 1]$, and replicated across three channels to yield $\tilde{\mathbf{x}}_d \in \mathbb{R}^{3 \times r \times r}$. We set $r = 512$, as this resolution maximises SD-VAE reconstruction fidelity across modalities (see Table I). The frozen encoder $\mathcal{E} : \mathbb{R}^{3 \times 512 \times 512} \rightarrow \mathbb{R}^{4 \times 64 \times 64}$ maps each slice to a latent representation

$$\mathbf{z}_d = \mathcal{E}(\tilde{\mathbf{x}}_d), \quad (1)$$

and the latent slices are stacked along the axial dimension to form the volumetric latent tensor

$$\mathbf{Z} = \text{stack}_{d=1}^D(\mathbf{z}_d), \quad \mathbf{Z} \in \mathbb{R}^{4 \times D \times 64 \times 64}. \quad (2)$$

This design removes the need for modality-specific autoencoder training and provides an $8 \times$ in-plane compression factor, compared with the $4 \times$ reduction commonly used in dedicated medical autoencoders.

C. Latent Flow Modeling and Generation

Once $\mathbf{Z}$ is defined by (2), generation reduces to a standard flow problem in $\mathbb{R}^{4 \times D \times 64 \times 64}$. A full-volume 3-D U-Net f_θ is trained directly on $\mathbf{Z}$ using the Just image Transformers (JiT) formulation [5], which predicts the clean latent and supervises in velocity space. This choice avoids the instability of ϵ -prediction in high-dimensional spaces and the blurriness of standard denoising diffusion probabilistic model (DDPM) x0-prediction, both of which are explained by [5].

For training, we form a noisy latent $\mathbf{Z}_t$ by linearly interpolating $\mathbf{Z}$ with Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathbf{Z}_t = (1 - t)\mathbf{Z} + t\epsilon, \quad t \sim \text{sigmoid}(\mathcal{N}(P_\mu, P_\sigma^2)). \quad (3)$$

The network f_θ predicts the clean latent $\hat{\mathbf{Z}} = f_\theta(\mathbf{Z}_t, t)$ and is optimized with the velocity-reweighted objective

$$\mathcal{L}_v = \mathbb{E}_{\mathbf{Z}, \epsilon, t} \left[\frac{\|\hat{\mathbf{Z}} - \mathbf{Z}\|_2^2}{(1 - t)^2} \right], \quad (4)$$

which is equivalent to supervising the predicted velocity $\hat{\mathbf{v}} = (\hat{\mathbf{Z}} - \mathbf{Z}_t)/(1 - t)$ against its target $\mathbf{v} = (\mathbf{Z} - \mathbf{Z}_t)/(1 - t)$, with

a $1/(1-t)^2$ factor that amplifies supervision at high noise levels.

At inference, generation starts from $\mathbf{Z}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and integrates the learned velocity field for K steps via an explicit solver (Euler or Heun) to obtain $\hat{\mathbf{Z}}$. Each latent slice $\hat{z}_d$ is decoded via (1) as

$$\hat{\mathbf{x}}_d = \mathcal{R}_r^\top(\mathcal{D}(\hat{z}_d)), \quad (5)$$

where $\mathcal{D} : \mathbb{R}^{4 \times 64 \times 64} \rightarrow \mathbb{R}^{3 \times H \times W}$ is the frozen decoder and $\mathcal{R}_r^\top$ restores the native in-plane resolution. The decoded slices are assembled into the output volume $\hat{\mathbf{x}} \in \mathbb{R}^{D \times H \times W}$.

D. Dataset

The datasets contained approximately 300 CT volumes from LIDC-IDRI and lymph-node data, 900 PET volumes from AutoPET, and 500 MRI volumes from IXI. All volumes were mapped to $[-1, 1]$ before latent encoding: CT intensities were clipped to $[-1000, 1500]$ HU, PET SUV values were clipped and transformed with a cubic-root mapping, and MRI volumes were normalized per volume using the 1st and 99th percentiles. CT, PET, and MRI were processed at 512^3 , 400^3 , and 256^3 , respectively. For JiT timestep sampling, we used $t \sim \sigma(\mathcal{N}(P_\mu, P_\sigma^2))$ with $P_\mu = -1.2$ and $P_\sigma = 1.2$.

III. RESULTS AND DISCUSSION

Table I demonstrates that the frozen SD-VAE achieves high reconstruction fidelity across MRI, CT, and PET, confirming that a pretrained image autoencoder can be used as an effective ready-to-use encoder for medical volumes. The best PSNR and MS-SSIM values are obtained at 512^2 for all modalities, showing that this resolution better preserves anatomical and functional structures. Based on these results, we use 512^2 slices for the latent generative experiments, without training modality-specific medical autoencoders.

TABLE I: Frozen SD-VAE slice-wise fidelity across input resolutions.

Res.	MRI		CT		PET	
	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM
256^2	27.72	0.968	42.10	0.985	43.32	0.978
400^2	31.94	0.990	42.61	0.994	42.25	0.991
512^2	34.08	0.995	44.00	0.996	44.35	0.996

We then evaluated full-volume generation using MONAI medical FID on 40 generated volumes per modality, computed independently on axial, coronal, and sagittal views. As shown in Table II, MRI and CT obtain the best average FID values, with CT being the most balanced across planes. PET shows good coronal and sagittal scores but a higher axial FID, which may reflect both axial-view inconsistencies and a possible mismatch between PET uptake patterns and the feature extractor used for FID. Therefore, we report these metrics together with qualitative generated volumes in Fig. 1.

TABLE II: MONAI medical FID per modality. Lower is better.

Modality	Shape	Avg. FID	Axial	Coronal	Sagittal
MRI	256^3	0.396	0.579	0.326	0.283
PET	400^3	0.594	1.258	0.259	0.266
CT	512^3	0.400	0.415	0.370	0.415

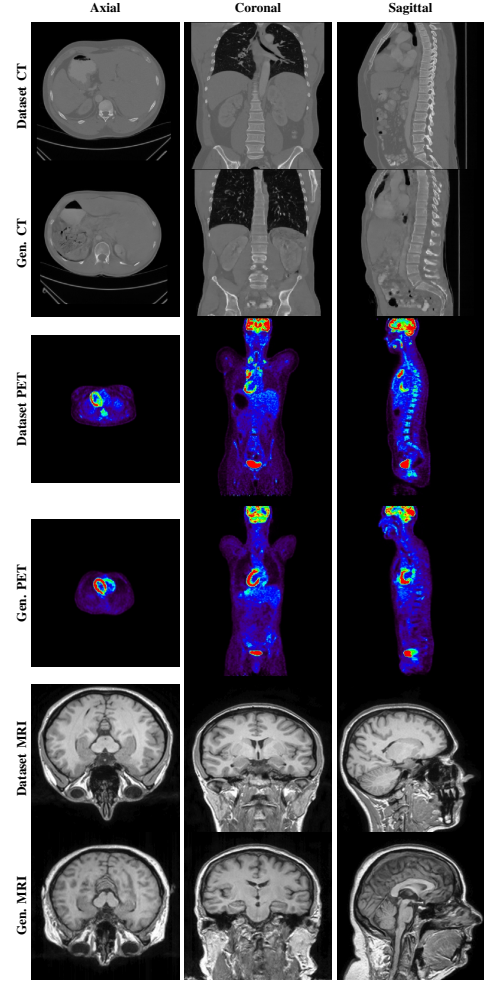


Fig. 1: Central axial, coronal, and sagittal slices of dataset and generated volumes for CT, PET, and MRI.

IV. CONCLUSION

We presented a scalable latent framework for full-volume 3-D medical image generation using a frozen SD-VAE pretrained on natural images as a ready-to-use slice-wise encoder, requiring no medical retraining or modality-specific adaptation. We validated that stacked slice latents support full-volume generative modeling across CT, PET, and MRI on a single consumer-grade GPU, producing coherent 3-D samples with strong Fréchet inception distance (FID) scores across all three orthogonal planes. Future work will exploit the learned volumetric prior for posterior sampling in 3-D inverse problems.

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